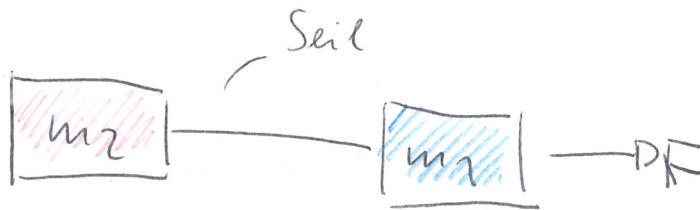


Zusatzaufgaben Dynamik

1)



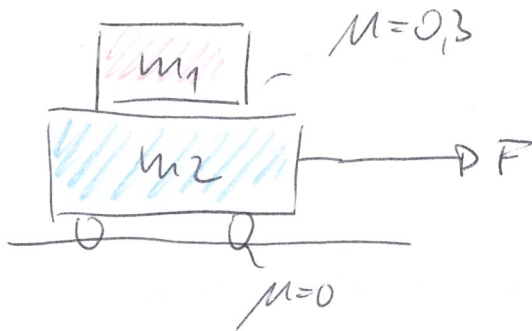
ges: Seilkraft

$$F = 200 \text{ N}$$

$$m_1 = 5 \text{ kg}$$

$$m_2 = 10 \text{ kg}$$

2)

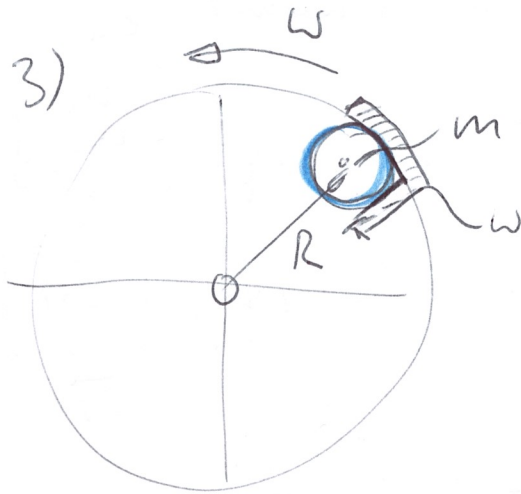


$$m_1 = 5 \text{ kg}$$

$$m_2 = 10 \text{ kg}$$

ges: Finne sodass m_1 nicht rutscht

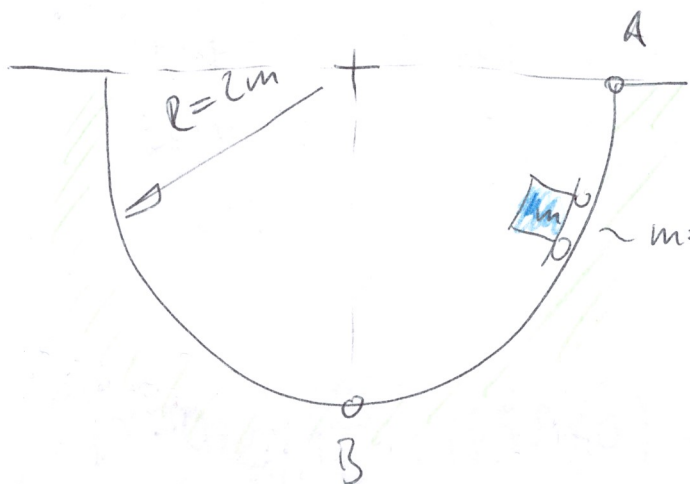
3)



es soll ω so berechnet werden, dass die Kugel in ω gegen den Winkel ω drückt

$$m = 1 \text{ kg}; R = 0.3 \text{ m}$$

4)



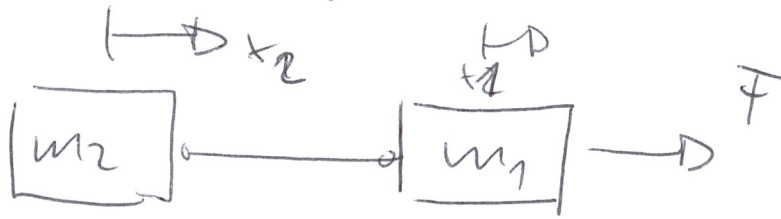
- Ein Skateboard fährt in A um

- ges die Geschw in B

- die Kraft in B

Dynamik Beispiel

1) Lineare Bewegung



$$F = 200 \text{ N}$$

$$m_1 = 5 \text{ kg}$$

$$m_2 = 10 \text{ kg}$$

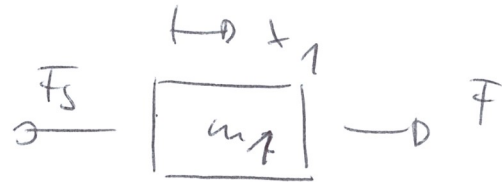
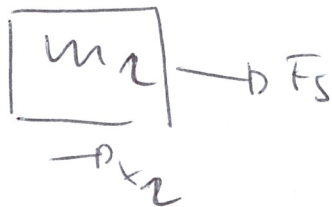
Formel: $m \cdot \ddot{x} = \sum \vec{F}$

gfs F_s

Freiheitsgrad: 1 Richtung linear; 2 Körper

Bedingung: $x_1 = x_2 \Rightarrow \ddot{x}_1 = \ddot{x}_2$

Körper + Freischnitten



$$m_2 \cdot \ddot{x}_2 = F_s$$

$$m_1 \cdot \ddot{x}_1 = -F_s + F$$

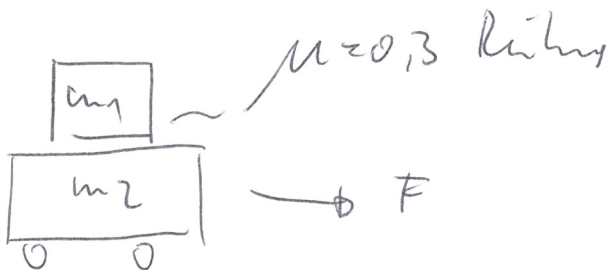
mit $\ddot{x}_1 = \ddot{x}_2 = \ddot{x}$

$$m_1 \cdot \ddot{x}_1 = m_2 \cdot \ddot{x}_2 + F \Rightarrow (m_1 + m_2) \ddot{x} = F$$

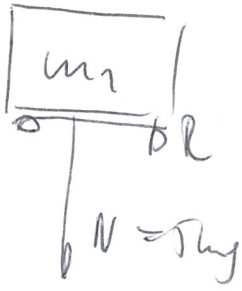
$$\ddot{x} = \frac{F}{m_1 + m_2} \rightarrow \frac{200}{15} = 13.3 \frac{\text{m}}{\text{s}^2} \quad F_s = m_2 \cdot \ddot{x} = 133 \text{ N}$$

2)

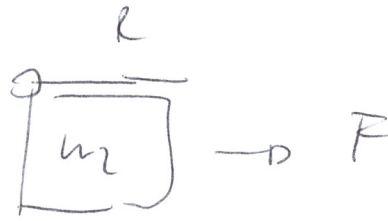
gfs Fmax



$$(m_1 + m_2) \ddot{x} = F \quad 5 \text{ kg} \quad 10 \text{ kg}$$



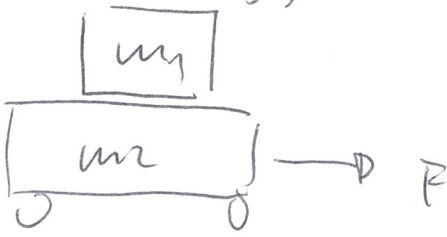
$$R = \mu \cdot N = 0,3 \cdot 50 = 15 \text{ N}$$



$$m_2 \cdot \ddot{x} = \sum F = -R + F$$

2)

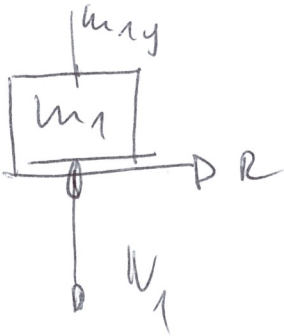
Slip



1. Freischnittsproblem i. Linie

$$m \cdot \ddot{x} = \sum \vec{F}$$

$$\ddot{y} = 0$$

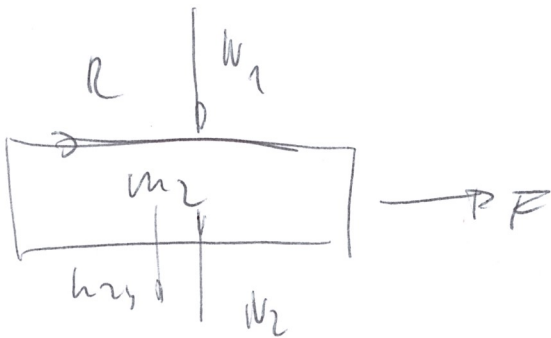


$$0 = -m_1 g + N_1 \rightarrow N_1 = m_1 \cdot g$$

$$m_1 \cdot \ddot{x} = R$$

$$R = \mu m_1 \cdot g$$

$$\ddot{x}_{\text{max}} = \mu \cdot g$$



$$0 = -N_1 + N_2 - m_2 g$$

$$m_2 \cdot \ddot{x} = -R + F$$

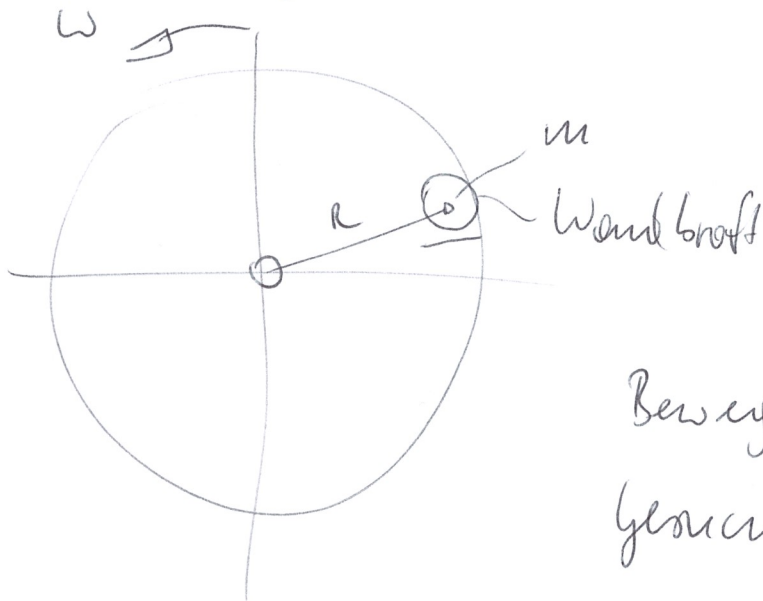
$$m_2 \cdot \ddot{x} + m_1 \cdot \ddot{x} = F$$

$$\cancel{m_2 \cdot \ddot{x}} + \mu m_1 g = F$$

$$(m_2 + m_1) \cdot \ddot{x} = F$$

$$F = (m_1 + m_2) \mu g //$$

3) Drehbewegung



$$m = 1 \text{ kg}$$

$$r = 0,3 \text{ m}$$

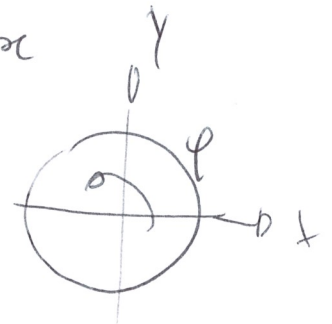
ges: ω , dass

Wandbohrer 100

Bewegungsart: Drehung
Gesucht: ω

Freiheitsgrade: φ um die z-Achse

Newton: $m \cdot \ddot{\vec{x}} = \vec{F}$



$$\ddot{\vec{x}} = (\ddot{r} - r\dot{\varphi}^2) \begin{bmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{bmatrix} + (r\ddot{\varphi} + 2\dot{r}\dot{\varphi}) \begin{bmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{bmatrix}$$

$\vec{e}_r$ $\vec{e}_\varphi$

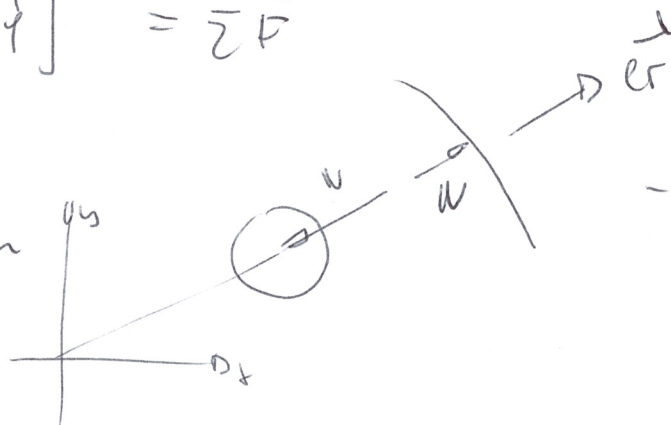
$r = \text{konst} \rightarrow \ddot{r} = 0$

$\omega = \text{konst} \rightarrow \ddot{\varphi} = 0, \dot{\varphi} = \text{konst}$

$$-m \cdot r \dot{\varphi}^2 \begin{bmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{bmatrix} = \vec{F}$$

$\vec{e}_r$

Frescheiden



$$-N \cdot \vec{e}_r = \vec{F}$$

$$-m r \dot{\varphi}^2 \cdot \vec{e}_r = -N \cdot \vec{e}_r$$

$\rightarrow$ nur Kräfte in radialer Richtung

$$m r \dot{\varphi}^2 = N$$

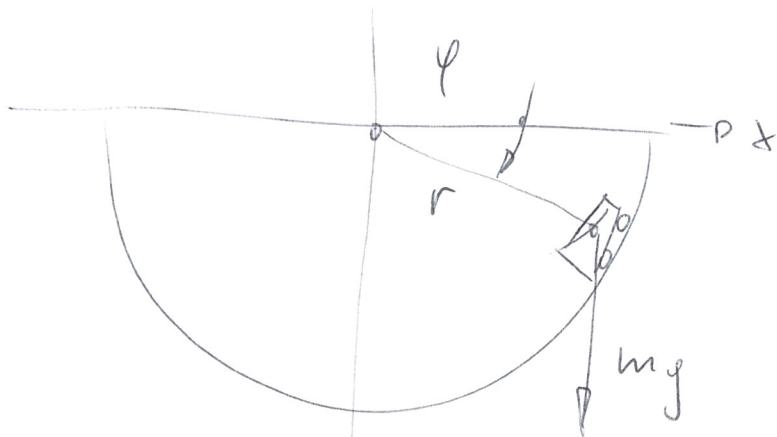
$$\dot{\varphi} = \sqrt{\frac{N}{m r}} \rightarrow \sqrt{\frac{100 \text{ N}}{1 \text{ kg} \cdot 0,3 \text{ m}}} = 5,8 \frac{1}{\text{s}}$$

4) Halfpipe

15

Stoßboarde köht bei $\varphi=0$ los

$r = \text{const}$



→ Drehung um φ (Z-Achse)

1 Freiheitsgrad

→ ges. Kräfte, $\ddot{\varphi}(\varphi)$

Formeln:

$$m \cdot \ddot{\vec{x}} = \sum \vec{F}$$

$$\ddot{\vec{x}} = (\ddot{r} - \dot{\varphi}^2 r) \begin{bmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{bmatrix} + (r - r \dot{\varphi}^2) \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix}$$

$\vec{e}_t$ $\vec{e}_r$

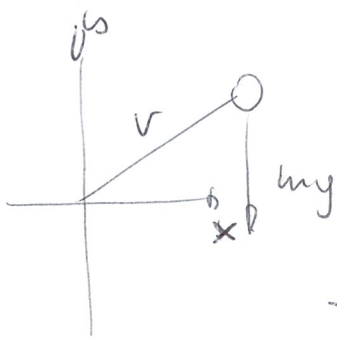
→ Kräfte in x, y, z -System

$r = \text{konstant}$ $\ddot{r}; \ddot{\varphi} = 0$; $\vec{e}_r$ Komponenten erzeugen kein Moment

$$\vec{x} = r \cdot \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} \rightarrow \text{Ruhelage in ersten Quadranten}$$

$$r \cdot \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} \otimes m \ddot{\varphi} \begin{bmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{bmatrix} = \sum \vec{M} \rightarrow m r^2 \ddot{\varphi} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \sum \vec{M}$$

$\sum \vec{M} \dots ?$ Freischnitten



$$\sum \vec{M} = \vec{r} \otimes \vec{F} \rightarrow r \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ -mg \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -mg \cos \varphi \end{bmatrix}$$

$$m r^2 \ddot{\varphi} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = -mg \cos \varphi \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\rightarrow \cancel{m} r \ddot{\varphi} = -g \cos \varphi \Rightarrow \ddot{\varphi} = -\frac{g}{r} \cos \varphi$$

$$\ddot{\varphi} = \frac{g}{r} \cdot \cos \varphi$$

$$\text{gib } \dot{\varphi}(\varphi)$$

$$\dot{\varphi} = \frac{d\dot{\varphi}}{dt} = \frac{d\dot{\varphi}}{d\varphi} \frac{d\varphi}{dt} = \frac{d\dot{\varphi}}{d\varphi} \cdot \dot{\varphi}$$

$$\frac{d\dot{\varphi}}{d\varphi} \cdot \dot{\varphi} = -\frac{g}{r} \cdot \cos \varphi \quad \rightarrow \quad \dot{\varphi} \cdot d\dot{\varphi} = -\frac{g}{r} \cdot \cos \varphi \cdot d\varphi$$

$$\frac{1}{2} \dot{\varphi}^2 = -\frac{g}{r} \cdot \sin \varphi + C \quad \dot{\varphi}_{\varphi=0} = 0 \quad C=0$$

$$\dot{\varphi} = \sqrt{-2 \cdot \frac{g}{r} \cdot \sin \varphi}$$

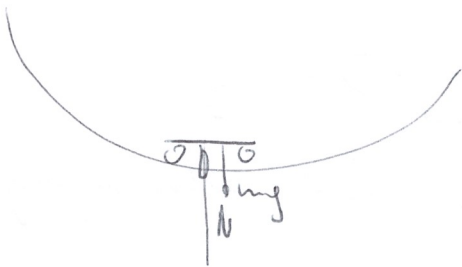
$$\varphi = -90^\circ = -\frac{\pi}{2}$$

$$\rightarrow \sin -\frac{\pi}{2} = -1$$

$$\dot{\varphi} = \sqrt{2 \frac{g}{r}}$$

Kraft auf das Stativ $\sin -\frac{\pi}{2}$

$$m \cdot \ddot{x} = \vec{z} \cdot \vec{F}$$



$$-m \cdot r \dot{\varphi}^2 \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -mg \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ N \\ 0 \end{bmatrix} \quad \varphi = -\frac{\pi}{2}$$

$$-m r \dot{\varphi}^2 \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ mg \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ N \\ 0 \end{bmatrix} \rightarrow m r \dot{\varphi}^2 + mg = N$$

$$m \cdot x \cdot 2 \frac{g}{r} + mg = N = D$$

$$3mg = N$$